Functional hard-sphere fluctuations on a regular Boltzmann lifespan

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October 14, 2026

Abstract

We prove a path-space upgrade of the finite-dimensional hard-sphere fluctuation theorem on an entire supplied regular Boltzmann interval. The setting is the deterministic, nonequilibrium, grand-canonical gas in three-dimensional whole space, with exact microscopic centering and no time smoothing. Relative to the pinned finite-history interfaces, one collective time translation of a connected history gives a shrinking-window factor at its sharp factorial-cumulant scale. This also localizes redundant contacts, whose times are determined rather than independent coordinates. A prefix-measurable global chronology indicator is retained through the translation and the flux-tree estimate. Initial-link cylinders preserve the connected activity scale, and a fixed Gaussian reserve keeps summability independent of the later moment order. A decorated final-slab output kernel controls the number of jumping particles. Third moments at scale $\mu^{-3/2}$, dyadic chaining, and a uniform exact-centering comparison yield J_1 convergence and C-tightness for every finite vector of compact smooth tests, with the Gaussian covariance of the credited finite-dimensional theorem.

1 The gas, the target, and the result

We work on $\mathbb{R}^3_x \times \mathbb{R}^3_v$, with $z = (x, v)$. All surface integrals on S^2 use ordinary, unnormalized area measure. For $\beta > 0$ put

$$\|g\|_{\mathrm{Bol},\beta} = \sum_{a\in\mathbb{Z}^3} \sup_{\substack{|x-a|\le 1\\ v\in\mathbb{R}^3}} e^{\beta|v|^2}|g(x,v)|. \tag{1.1}$$

For relative velocity $w = v - v_*$ and $\omega \in S^2$ use

$$B(w,\omega) = [w\cdot\omega]_+, \qquad v' = v - (w\cdot\omega)\omega, \qquad v'_* = v_* + (w\cdot\omega)\omega. \tag{1.2}$$

The Boltzmann collision operator is

$$Q(f,f)(x,v) = \int_{\mathbb{R}^3\times S^2} B(v-v_*,\omega)(f'f'_* - ff_*)\,dv_*\,d\omega,$$

with all four factors evaluated at the same position x.

Assumption 1.1 (The kinetic hypothesis)**.** *The function $f_0 \ge 0$ is a smooth probability density and, for some $\beta > 0$,*

$$\|f_0\|_{\mathrm{Bol},2\beta} + \|\nabla_x f_0\|_{\mathrm{Bol},2\beta} < \infty.$$

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There is a nonnegative classical solution of $(\partial_t + v\cdot\nabla_x)f = Q(f,f)$*, with initial datum* f_0*, on a prescribed finite interval* $[0,T]$*,* $T>0$*, such that*

$$\sup_{0\le t\le T}\sup_{x,v} e^{2\beta|v|^2} f_t(x,v) < \infty. \tag{1.3}$$

This is exactly Assumption `ass:kinetic` of the pinned source [1]. No smallness or global existence assumption is added.

For sphere diameter $\varepsilon>0$ set $\mu=\varepsilon^{-2}$ and

$$\mathcal{D}_N^\varepsilon = \{Z_N : |x_i - x_j| > \varepsilon \text{ for } i\neq j\}.$$

The initial law is

$$\mathbb{P}_\varepsilon(N, dZ_N) = \mathcal{Z}_\varepsilon^{-1}\frac{\mu^N}{N!}\mathbf{1}_{\mathcal{D}_N^\varepsilon}(Z_N)\prod_{i=1}^N f_0(z_i)\,dZ_N, \qquad N=0,1,2,\dots. \tag{1.4}$$

The empty configuration is included. From the sampled initial configuration, the particles move by deterministic free flight and elastic hard-sphere collisions. We use the almost-everywhere flow used in [1], and adopt right-continuous velocities (post-collision values at collision times) as our convention. For each fixed particle number N, the initial law has a Lebesgue density on $\mathcal{D}_N^\varepsilon$. At any prescribed time the contact configurations lie on a finite union of codimension-one surfaces $|x_i-x_j|=\varepsilon$, of Lebesgue measure zero. The measure-preserving hard-sphere flow therefore makes their preimages null, outside its already null exceptional set. Conditioning on N and summing over its countably many values (with no contact for $N=0,1$) proves that a collision at a fixed time has probability zero. Thus this convention leaves all fixed-time inputs from H unchanged. There is no random collision rule or physical resampling.

For real $\phi\in C_c^\infty(\mathbb{R}^6)$ define the unsmoothed fluctuation

$$X_t^\varepsilon(\phi) = \mu^{-1/2}\left(\sum_{i=1}^N \phi(z_i(t)) - \mathbb{E}\sum_{i=1}^N \phi(z_i(t))\right). \tag{1.5}$$

The centering is the exact microscopic expectation at each time. It is not replaced by $\int f_t\phi$.

Write

$$\Delta\phi = \phi(x,v') + \phi(x,v_*') - \phi(x,v) - \phi(x,v_*).$$

The Gaussian target has weak drift and noise form

$$A_t\phi = v\cdot\nabla_x\phi + \int B(v-v_*,\omega) f_t(x,v_*)\Delta\phi\,dv_*\,d\omega, \tag{1.6}$$

$$C_t(\phi,\psi) = \frac{1}{2}\int B(v-v_*,\omega) f_t(x,v) f_t(x,v_*)\Delta\phi\Delta\psi\,dx\,dv\,dv_*\,d\omega. \tag{1.7}$$

Let $U(r,t)$ be the backward propagator of A_t constructed in [1, Section 2, `lem:target-propagator`]. The target is the centered jointly Gaussian field with covariance

$$\begin{aligned} K_{\phi,\psi}(t,s) = &\int f_0(z)U(0,t)\phi(z)U(0,s)\psi(z)\,dz \\ &+ \int_0^{t\wedge s} C_r\big(U(r,t)\phi, U(r,s)\psi\big)\,dr. \end{aligned} \tag{1.8}$$

Its initial covariance is $\int f_0\phi\psi$, with no fixed-number subtraction. Equivalently it solves $d\zeta_t(\phi) = \zeta_t(A_t\phi)dt + dW_t(\phi)$ with the independent initial field and Gaussian noise prescribed by (1.7).

Theorem 1.2. *Under Assumption 1.1, relative to the source interfaces stated in Section 2, for every finite list of real* $\phi_1, \dots, \phi_d \in C_c^\infty(\mathbb{R}^6)$,

$$(X_\cdot^\varepsilon(\phi_1), \dots, X_\cdot^\varepsilon(\phi_d)) \implies (\zeta_\cdot(\phi_1), \dots, \zeta_\cdot(\phi_d))$$

in $D([0,T], \mathbb{R}^d)$ *with the Skorokhod* J_1 *topology. The sequence is C-tight, and the target is the continuous version of the Gaussian field with covariance* (1.8).

The interval is the whole supplied $[0,T]$. The theorem concerns the actual gas (1.4), not an independently restarted gas or an integrated observable. We make no weak-convergence assertion for the uniform-topology Borel space on all of D.

Bodineau–Gallagher–Saint-Raymond–Simonella proved short-time nonequilibrium process convergence for a periodic gas [2], and long-time equilibrium process convergence by an invariant-measure argument [3]. Deng–Hani–Ma supply the regular-lifespan framework [4]; their Section 1.4.3 identifies fluctuations as a further direction. The pinned source [1] gives the finite-dimensional law on the regular lifespan and explicitly leaves path-space tightness outside its conclusion. The new operation here is the collective shrinking-window estimate in Section 4. The moment, continuity, chaining, and centering arguments are credited separately. These comparisons do not assert worldwide priority.

2 The credited finite-history interfaces

Throughout, H denotes [1] at the immutable revision

`fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`.

The truth of H's theorems is stipulated for this work. We state their operational interfaces to distinguish them from the new deductions. The shorthands H01–H08 name the following pinned source files, whose actual section numbers follow the inclusion order in H's main file:

Shorthand	Section of H	File under `build/sections/`
H01	1	`01-introduction.tex`
H02	2	`02-initial-target.tex`
H03	3	`03-dhm-inputs.tex`
H04	4	`04-cluster-expansion.tex`
H05	7	`05-cutoff.tex`
H06	5	`06-sharp-asymptotics.tex`
H07	6	`07-gaussian-limit.tex`
H08	8	`08-transfer.tex`

Their titles, respectively, are "Introduction and statement of the theorem", "The initial ensemble and the Gaussian target", "Layered histories and the analytic inputs", "A componentwise expansion with particle records", "The probability of a dynamical cutoff", "Sharp bounds and the limiting tree measures", "Identification of the covariance and the Gaussian limit", and "Exact centering and the hard-sphere fluctuation limit". Typewriter theorem and equation labels below are exact labels in these pinned files.

2.1 One auxiliary dynamics and factorial normalization

Fix the dimensional collision-excess cap Γ first and then a large fixed number L of slabs, with $\tau = T/L$. H uses the size cutoffs

$$A_L = |\log \varepsilon|, \qquad \Lambda_i = A_i^{30}, \qquad A_{i-1} = \Lambda_i^{30}, \tag{2.1}$$

with harmless integer rounding. Within each slab an incoming collision is accepted if adjoining it to the current collision graph leaves its component with at most Λ_i labels and excess at most Γ; otherwise both particles continue straight. At an interface only this collision prehistory is reset. Phase points and records are pasted without a jump. This is H's single auxiliary pasted dynamics; the target physical flow is not reset. It preserves N and has finitely many accepted collisions. We put hats on its fields and center them by their own exact expectations.

If $F^{\varepsilon}_{h,t}$ is its ordered factorial measure normalized by μ^h, then, for a nonnegative measurable test H_0,

$$F^{\varepsilon}_{h,t}(H_0) = \mu^{-h}\mathbb{E} \sum_{i_1,\dots,i_h \text{ distinct}} H_0(z_{i_1}(t),\dots,z_{i_h}(t)).$$

The same definition allows specified records on the designated labels. Normalized factorial cumulants satisfy

$$F^{\varepsilon}_{h,t} = \sum_{\pi\in\mathcal{P}([h])} \bigotimes_{V\in\pi} c^{\varepsilon}_{|V|,t}. \tag{2.2}$$

H02, Proposition `prop:initial`, gives

$$F^{\varepsilon}_{h,0} \le f_0^{\otimes h}, \qquad \mathbb{E}(N)_h \le \mu^h, \qquad \sup_{\varepsilon\le\varepsilon_0} \mathbb{E}(N/\mu)^p < \infty \quad \text{for every fixed integer } p. \tag{2.3}$$

The factorial-mass bound persists under the number-preserving dynamics.

A scalar record is initially zero and is unchanged by collisions. It has updates $y \mapsto y + \phi_j(z)$ at prescribed pulse times. Its sum records the corresponding multitime empirical sum. H07, `eq:limit-factorial-ordinary`, gives for $S = \sum_i y_i(T)$

$$\kappa_j(S) = \sum_{\pi\in\mathcal{P}([j])} \mu^{|\pi|} \int c^{\varepsilon}_{|\pi|,T}(d\xi) \prod_{V\in\pi} y_V^{|V|}. \tag{2.4}$$

The blocks V are associated with distinct particle slots in a fixed order. This is a finite formal identity through order j: expand $\prod_i[1 + (e^{uy_i} - 1)]$ and take its logarithm. Each factor receives a nonempty set of differentiations, giving the displayed partitions. No uniform radius of exponential integrability is needed.

2.2 The exact one-slab root-output kernel

Here we record the finite identity used later for a nonadditive flag. For a finite particle set V_0 let Ψ_C be the autonomous capped history of $C \subset V_0$, and let κ_C indicate that its accepted collision graph is connected. Singletons have $\kappa_C = 1$. For disjoint autonomous clusters let $a(C, D)$ indicate the existence of an incoming cross-contact that would be accepted with their prefix prehistories, and set $\chi(C, D) = 1 - a(C, D)$. H04, Lemma `lem:autonomous-partition`, states, pointwise on its common full-measure set,

$$H_0(\Psi_{V_0}|_S) = \sum_{\mathcal{C}\in\mathcal{P}(V_0)} \prod_{C\in\mathcal{C}} \kappa_C \prod_{\{C,D\}\subset\mathcal{C}} \chi(C, D) H_0(\Psi_{\mathcal{C}}|_S) \tag{2.5}$$

for every bounded measurable test of the root histories. The selected juxtaposition is the actual capped history; no independence of input is assumed.

For a cluster block V_c write $u(V_c) = \sum_{G \text{ connected on } V_c} \prod_{e\in G}(-a_e)$ and $u(\{C\}) = 1$. The rooted forest exploration stops when a rooted cluster component reaches $M_i = \lfloor \Lambda_i/2 \rfloor$. Its retained root-output kernel, for roots $S = [s]$ and reached labels $V_0 = [s + j]$, is

$$\mathcal{K}_{S,V_0}(Z, dZ'_S) = \sum_{\mathcal{C}\in\mathcal{P}(V_0)} \prod_{C\in\mathcal{C}} \kappa_C \sum_{\substack{\mathcal{B}\in\mathcal{P}(\mathcal{C}) \\ (\cup_{C\in V_c} C)\cap S\neq\varnothing,\ |V_c|<M_i}} \prod_{V_c\in\mathcal{B}} u(V_c)\delta_{\Psi_{\mathcal{C}}(Z)|_S}(dZ'_S). \tag{2.6}$$

The inner restrictions hold for every block. The corresponding operator is

$$\mathcal{L}_{i,s}F = \sum_j \frac{\mu^j}{j!} \int F_{s+j}(dZ)\mathcal{K}_{S,V_0}(Z,\cdot). \tag{2.7}$$

The sum is finite under the caps. H04's one-slab identity is $F'_s = \mathcal{L}_{i,s}F + R_s$, with $|R_s(H_0)| \le (\mathcal{E}_{i,s}F)(|H_0|)$, where $\mathcal{E}$ is a positive stopped-forest kernel with the same output map. Each forest exploration branch has multiplicity at most one. Spectator clusters retain mutual compatibility and cancel by (2.5) before integration. The fresh-label identity is

$$\frac{1}{(n-s)!}\binom{n-s}{j} = \frac{1}{j!(n-s-j)!}. \tag{2.8}$$

It yields exactly the coefficient in (2.7). This kernel identity, including the output before pushforward, is an imported interface.

2.3 Resummed diagrams and errors

H04 defines ordinary independent one-root inputs g_i by the entire retained one-root map on independent inputs, and proves their Gaussian Bol envelopes and convergence to $f_{i\tau}$, including partial slabs. Required record lines continue through all interfaces to zero. Other one-root blocks attached through only one line can be resummed into the ordinary g_i. This is an exact finite regrouping, not a factorization of the true interface law. Forgetting a record commutes with the finite hierarchy and the regrouping.

The retained hierarchy factors over its global components. Finite inversion of (2.2) therefore keeps exactly the globally connected diagrams; initial cumulant blocks count as connections, artificial dummy links do not. H04, Theorem `thm:marked-expansion`, supplies for fixed root order h and a fixed record list bounded in number by r_{rec}:

1. an unweighted total-variation error at most $C_h e^{-c\Lambda_L}$ between the actual cumulant and the retained connected sum;
2. for every assigned power $q > 0$, an auxiliary K after that order and list such that the retained complexity tail $\rho > K$ has absolute variation at most ε^q;
3. the original fresh-label coefficients, with denominator $\prod_i j_i!$, and at bounded complexity

$$n_i - j_i = O_{K,h,r_{\mathrm{rec}},L}(1), \qquad \prod_i \frac{1}{j_i!} \le \frac{(1+\sum_i n_i)^{C_{K,h,r_{\mathrm{rec}},L}}}{\prod_i n_i!}. \tag{2.9}$$

All constants in an exponential base per ordinary atom are independent of K and of the subsequently fixed moment order. Distinguished choices cost fixed polynomials; in the coarse regime they cost cutoff powers. These claims follow from the actual finite kernels and positive stopped bounds, not from a postulated uniform limit theorem.

The coarse tail has the form

$$C_{h,r_{\mathrm{rec}}}\varepsilon^{-C_* h - C_{r_{\mathrm{rec}}} + c_* K/2}, \tag{2.10}$$

where r_{rec} is the fixed record bound, corresponding to H04's parameter m. H03's allowed boundary modifications include a top-root deficit $\varepsilon^{-C_* h}$ and at most $\varepsilon^{-q_0/81}$ for q_0 specified lines requiring dummy initial continuations. The auxiliary dummy straight roots have unit-mass Gaussian Bol inputs; their activities and root normalizations cancel. They are used only for coarse domination

and removed before the bounded-complexity physical-link analysis. Their number is fixed by the root/record list, not by the number of collisions. All stopped bounds are uniform on a partial final slab and assert no Gaussian weight for a general TV error.

2.4 Physical charts and sharp summation

A C-atom scatters a pair elastically; an O-atom records an incoming encounter with straight continuation. On replacing bounded-order initial cumulants by H06's forest domination, each initial edge has weight

$$\mathbf{1}_{|x_p(0)-x_q(0)|\le R_{K,D}\varepsilon}+\varepsilon^D. \tag{2.11}$$

There are at most K such edges. They join lines born at zero. First retain only their physical indicator terms. For a graph with m distinct birth lines, c global components, p physical initial edges, a_i dynamic mergers and n_i lines in slab i, H06 gives

$$a:=\sum_i a_i=m-c-p,\qquad \sum_i(n_i-a_i)=O_K(1),\qquad 0\le a_i\le n_i. \tag{2.12}$$

The number of redundant contacts is $O_K(1)$. A continued physical line is one vertex and has no new birth variable at an interface.

H06, Lemma `lem:sharp-merger-coordinates`, is a change of variables on each prescribed physical incoming straight-segment branch. Its coordinates are independent birth velocities, one common translation per global component, the relative affine origins of physical initial forest edges, and (t_g,ω_g) at every dynamic merger. Its merger Jacobian is

$$\varepsilon^2[(v-v_*)\cdot\omega_g]_+dt_g\,d\omega_g. \tag{2.13}$$

A redundant contact has no free parameter: for a relative straight flight $r(t)=d+tu$ on the common segment interval I,

$$\int_I\delta(|r(t)|-\varepsilon)[-u\cdot r(t)/\varepsilon]_+dt=\mathbf{1}_{\{\text{an entering intersection occurs in } I\}}. \tag{2.14}$$

There is at most one entering root on the whole relative flight. Merger coordinates have multiplicity one on the specified physical branch. Their activity is μ^{m-h}; the initial forest solve has determinant one.

The source fresh-label and energy-bin summation gives, at fixed complexity,

$$C_K(1+n)^{\gamma_K}(C\tau\sqrt{L})^n\mu^{1-h},\qquad n=\sum_i n_i, \tag{2.15}$$

for connected absolute diagram measures. The original physical link volume $O(\varepsilon^3)$ supplies an extra ε per initial link. Erasing a tail edge in (2.11) increases the component count by at most one; erasing at most K costs at most ε^{-2K}. Thus $D>2K+q_{\mathrm{tail}}$ makes all terms with an initial tail $O(\varepsilon^{q_{\mathrm{tail}}}\mu^{1-h})$, where $q_{\mathrm{tail}}>0$ is the assigned tail accuracy, distinct from M_i and M_ϕ. We will reproduce the summation steps needed after the new enlargement, rather than assume (2.15) continues to hold there.

2.5 The Gaussian field, finite-dimensional limit, and coupling

H02 gives the target covariance (1.8), its weak identity on tests with small Gaussian growth, and the bounded field maps described in Section 8. H01, Theorem `thm:main`, gives joint finite-dimensional

convergence for every fixed finite time/test list. H05, Proposition `prop:cutoff`, gives, for every sufficiently large fixed L, a coupling with the same initial configuration such that

$$\mathbb{P}(E_\varepsilon) \le C\varepsilon^{1+\eta_0}, \qquad \eta_0 = 1/20, \tag{2.16}$$

where E_ε is any true/pasted discrepancy at any time in $[0,T]$. This is a simultaneous path comparison, not a moment estimate on the exceptional event. These interfaces, (2.3), and the history package above are the imported results used in the proof.

3 A fixed Gaussian reserve and one choice of slabs

The order of choices is essential. Let $\beta_* = \beta/4$, before any moment order or complexity threshold. H03's exponents satisfy $\beta_i \ge 9\beta/20 > \beta_*$. For each fixed candidate L and $\varepsilon \le \varepsilon_0(L)$, H04's one-root estimate gives

$$\|g_i - f_{i\tau}\|_{\mathrm{Bol},\beta_i} \le \varepsilon^{\theta_i} \le 1.$$

H03, Lemma `lem:kinetic-envelopes`, bounds $\sup_{t\le T}\|f_t\|_{\mathrm{Bol},\beta}$ in terms of the kinetic data. Consequently there is a constant B_0 depending only on those data, with

$$\sup_i \|g_i\|_{\mathrm{Bol},\beta_*} \le B_0 \quad (\varepsilon \le \varepsilon_0(L)). \tag{3.1}$$

The smallness threshold may depend on L; the envelope constant does not. This prevents a circular choice of L from an L-dependent atom base.

For birth velocities $v_1,\dots,v_m$ put $E_0 = \sum_p |v_p|^2$. Integrating one spatial translation by a spatial L^1 norm and the remaining inputs by their spatial suprema gives $C^m e^{-\beta_* E_0}$. Reserve a fraction now, independently of later K: any fixed-degree polynomial satisfies

$$(1+E_0)^{p/2} \le C_{p,\beta_*} e^{\beta_* E_0/4}. \tag{3.2}$$

The remaining $e^{-3\beta_* E_0/4}$ supplies both a velocity-volume integral and a dyadic suppression. For $E_0 \in [P/2, P]$,

$$e^{-3\beta_* E_0/4} \le e^{-\beta_* E_0/4} e^{-\beta_* P/4}. \tag{3.3}$$

The lowest bin $E_0 < 1$ is treated separately.

All per-birth, normal, energy-bin, and ordinary-atom constants determined by this reserve are fixed independently of K and the order. Choose L once large enough for the source coarse/cutoff thresholds at these exponents and for $CT/\sqrt{L} < 1$. Only after this choice specify an observation list, a moment order, an accuracy power, and K, D. H permits every sufficiently large fixed L and the entire supplied T is retained. No later increase of K or of a fixed moment order changes the auxiliary dynamics.

4 The collective contact-window estimate

For a retained connected diagram and a designated line ℓ, let $N_J(\ell)$ be the number of its C-atoms incident to ℓ with time in a bounded interval J. We use absolute birth inputs and retain the prescribed history domains. Bounded record multipliers can be replaced by their suprema.

Lemma 4.1 (Contact windows)**.** *For fixed root order h, fixed complexity bound K, fixed initial sharpness D, and a fixed bounded record list, the physical-link connected diagrams $\mathfrak{D}_K$ satisfy*

$$\sum_{\mathsf{D}\in\mathfrak{D}_K}\int N_J(\ell)\,d|\mathsf{D}| \le C\mu^{1-h}|J|. \tag{4.1}$$

The constant is uniform in sufficiently small ε, in J, and in all record or pulse positions. The counted C-atom may be a dynamic merger or a redundant contact.

We prove the lemma in several steps. No quantitative disappearance of cycles is used.

4.1 Take the physical chart before enlarging it

Fix a graph G, its C/O choices, slab assignments, prescribed global event order, physical initial forest, and merger/redundant classification. Choose one marked C-slot a_* incident to ℓ. The global order is the ordinary chronological refinement in the source enumeration; it adds no independent permutation factor. In the connected case the physical coordinates are

$$\xi = (v, X, y, (t_g, \omega_g)_g), \qquad d\xi = dv\,dX\,dy\,dt\,d\omega,$$

with $a = m - 1 - p$ dynamic mergers. Let j_i be the fresh labels in slab i. To display all discrete weights, write

$$\mathfrak{a}_G = \mathfrak{b}_G \prod_i \frac{1}{j_i!}, \tag{4.2}$$

where $\mathfrak{b}_G$ contains the fixed graph's forest-domination and other discrete coefficients. They will be summed with H's factorial count; none is hidden in a constant that discards the denominator. The physical change of variables gives

$$\int \mathbf{1}_{t_{a_*}\in J}\,d|\mathsf{D}_G| \le \mathfrak{a}_G\mu^{m-h}\varepsilon^{2(m-1-p)}\int F_G(\xi)\mathbf{1}_{t_{a_*}(\xi)\in J}\,d\xi. \tag{4.3}$$

Here F_G contains the birth factors, physical-link balls, merger fluxes, and all physical segment, lifetime, acceptance, slab, and order indicators. Equation (2.13) is used on this physical chart only. What follows is a positive majorant after that change, not a new global Jacobian claim.

Let A_p be the affine origin at zero of line p's pre-first-event segment, whose independent velocity is v_p. This need not be the eventual piecewise path's value at zero. For initial forest edges orient p_l to q_l and set $y_l = A_{q_l} - A_{p_l}$. Their endpoints are born at zero, so these are the actual initial position differences on the physical chart. The forest determines each initial group up to a common translation. A positive-interface birth has its own independent input; an interface continuation does not. At the end choose $X = A_{\mathrm{rep}}$ for a fixed representative.

4.2 A prefix-measurable chronological extension

Start all m lines on their pre-first affine segments on the whole real axis. For each group keep its old segments, current incoming affine segments, and one free common translation. Process the prescribed global slots:

1. At a merger between distinct groups, with prescribed t_g, ω_g, solve the source contact equation $X_1 - X_2 = -d(t_g) - \varepsilon\omega_g$. Translate every segment already constructed in the moved group by the same vector. All earlier internal contacts, initial y_l, times, and velocities are preserved. Keep one common translation for the joined group. Use the current incoming segments extrapolated to t_g and require positive incoming flux.

2. At a redundant slot use the two current segments. Take their unique transversal entering root at or after the later preceding incident event time; a missing preceding time means $-\infty$. Abort if no such root exists. This root and its normal are determined, with no extra variable.

3. In both cases require the new time to be strictly later than the preceding global slot; the first lower bound is $-\infty$. Abort otherwise. On success perform its elastic C-update or straight O-update and continue.

Write R_G^{ext} for the incoming/existence tests and Q_G for the strict global order tests. Their product is the product of prefix success indicators, extended by zero at the first failure. No future trajectory is needed on a failed branch. The forest solve and entering quadratic roots are measurable prefix operations. After fixing the final common X, the successful construction is unique and is a chronological all-time path.

There is one important distinction between physical null sets and extended rejection. If a redundant O-update is followed by another redundant slot on that same unchanged relative flight, the entering-root rule can return exactly the same time. This is a structural tie on the extended chart, not a null-set assertion. Strict Q_G rejects it with weight zero. Such a branch cannot describe two physical incoming contacts on a fixed segment pair. The tie and its rejection are shift invariant. Other physical grazing and simultaneous-contact exclusions retain the source's almost-everywhere convention.

All m lines in this extension are present at every real time. Elastic C-updates conserve pair energy and O-updates change none. Therefore

$$\sum_{p=1}^{m} |v_p^{\mathrm{current}}|^2 = E_0 \tag{4.4}$$

on every successful prefix. In particular any slab's subset has energy at most E_0, even when an extended event falls outside a physical lifetime.

The input factor remains $|g_{(p)}(x_p^{\xi}(b_p), v_p)|$ at its original fixed birth time b_p. On a nonphysical branch its independent second argument v_p may differ from the instantaneous velocity at b_p. This is allowed for a positive extension, not asserted to be a gas law. It agrees on the physical domain. Every spatial argument remains $X + \delta_p$, and its Gaussian argument is the independent pre-first velocity.

Define, on successful branches and by zero after failure,

$$\begin{aligned} F_G^{\mathrm{ext,Q}}(\xi) = \prod_p |g_{(p)}(x_p^{\xi}(b_p), v_p)| \prod_l \mathbf{1}_{|y_l| \le R_{K,D}\varepsilon} \\ \cdot \prod_g [(\Delta v_g) \cdot \omega_g]_+ \mathbf{1}_{t \in \mathcal{D}} R_G^{\mathrm{ext}} Q_G, \end{aligned} \tag{4.5}$$

where $\mathcal{D}$ prescribes the merger slabs and relative merger order. Records have already been bounded; they are not reinterpreted on this extension.

Lemma 4.2 (Physical domination)**.** *On the physical chart $F_G \le F_G^{\mathrm{ext,Q}}$ and the marked time agrees with its extended time.*

Proof. Induct on the slots. A physical merger solves the same relative-translation equation. A physical redundant contact lies on the current common segment interval, which is contained in the extension's forward ray. The straight relative flight has at most one entering root, so the physical root is the computed root. Every physical global order test succeeds. Thus the paths and velocities coincide throughout the physical lifetimes and $Q_G = 1$ there. Every omitted physical factor is between zero and one. □

For comparison, the local entering test alone is insufficient. With unit diameter and incoming collinear data there are merger slots at $1/5$ and $2/5$, then a determined redundant time $7/15$, then a merger prescribed at $9/20$. All local fluxes can be positive, but $7/15 > 9/20$. The construction above rejects the final slot. The accompanying exact counterchart documents this failure of the chronology-free extension; it is not a counterexample to the physical gas.

4.3 The collective shear

For $\theta \in \mathbb{R}$ define

$$\Phi_\theta(v, X, y, t, \omega) = \big(v, X - \theta v_{\mathrm{rep}}, (y_l - \theta(v_{q_l} - v_{p_l}))_l, t + \theta\mathbf{1}, \omega\big). \tag{4.6}$$

Lemma 4.3 (Covariance and measure)**.** *The shear preserves success or failure and its prefix, R_G^{ext}, Q_G, and all fluxes. On successful histories,*

$$x_p^{\Phi_\theta \xi}(s) = x_p^{\xi}(s - \theta) \quad \textit{for all } p, s.$$

Every computed event time increases by θ. The map preserves $dv\, dX\, dy\, dt\, d\omega$ and has inverse $\Phi_{-\theta}$.

Proof. Time translation changes each pre-first affine origin by $A_p \mapsto A_p - \theta v_p$, giving precisely the X, y shear. At a merger, translate all old segments of the moved group. Its contact equation at $t_g + \theta$ is the old equation at t_g, and the incoming sign is the same. At a redundant slot the relative flight and its preceding-time lower bound translate together, so its entering root and normal have the claimed covariance. The global order comparison uses two times both increased by θ. This proves covariance and matching aborts by prefix induction; structural ties also remain ties. Future group translations preserve the old internal conditions. Uniqueness of the final forest solve fixes X. The affine-origin argument does not use the generally false equality $x_p(-\theta) = A_p - \theta v_p$ for a full path that has already scattered. For fixed v the map translates (X, y, t). On the full parameter space it has block triangular derivative with identity diagonal, and leaves the sphere variables fixed. The inverse proves measure preservation. □

4.4 Average first, retaining chronology

Put $\Theta = \tau$. For the extended marked integral, change variables and average to obtain

$$\begin{aligned} I_G &:= \int F_G^{\mathrm{ext,Q}}(\xi) \mathbf{1}_{t_{a_*}^{\mathrm{ext}}(\xi) \in J}\, d\xi \\ &= \frac{1}{\Theta} \int_0^{\Theta} \int F_G^{\mathrm{ext,Q}}(\Phi_\theta \eta) \mathbf{1}_{t_{a_*}^{\mathrm{ext}}(\eta) + \theta \in J}\, d\eta\, d\theta. \end{aligned} \tag{4.7}$$

All integrands are nonnegative, so Tonelli applies. First integrate X. Its coefficient in every birth position is the identity, and

$$\int \prod_p |g_{(p)}(X + \delta_p, v_p)|\, dX \le \big\| g_{(p_0)}(\cdot, v_{p_0}) \big\|_1 \prod_{p \neq p_0} \big\| g_{(p)}(\cdot, v_p) \big\|_\infty =: \mathfrak{B}(v) \le C^m e^{-\beta_* E_0}. \tag{4.8}$$

The bound is independent of the displacements, including their dependence on the shift and on fixed birth times.

For $u_l = v_{q_l} - v_{p_l}$, a single simultaneous shift yields the containing cylinder

$$\begin{aligned} \mathbf{1}_{|y_l - \theta u_l| \le R_{K,D}\varepsilon} &\le \mathbf{1}_{y_l \in \mathrm{Cyl}_l}, \qquad \mathrm{Cyl}_l = [0, \Theta] u_l + B(0, R_{K,D}\varepsilon), \\ |\,\mathrm{Cyl}_l\,| &\le \pi R_{K,D}^2 \varepsilon^2 \Theta |u_l| + \frac{4\pi}{3} R_{K,D}^3 \varepsilon^3. \end{aligned} \tag{4.9}$$

Replacing simultaneous membership by the product of these containing sets is a positive upper bound; no separate shift is introduced per initial link. The time-window integral is at most

$$\int_0^\Theta \mathbf{1}_{t+\theta\mathbf{1}\in\mathcal{D}}\mathbf{1}_{t^{\mathrm{ext}}_{a_*}+\theta\in J}\,d\theta \le |J|\mathbf{1}_{t\in\mathcal{D}_\Theta}, \qquad \mathcal{D}_\Theta = \mathcal{D} - [0,\Theta]\mathbf{1}.$$

The fluxes and both prefix indicators are shift invariant. Thus

$$I_G \le \frac{|J|}{\Theta}\int dv\,\mathfrak{B}(v)\int dy\prod_l \mathbf{1}_{y_l\in \mathrm{Cyl}_l}\int_{\mathcal{D}_\Theta} dt\int d\omega\, R_G^{\mathrm{ext}}Q_G\prod_g[(\Delta v_g)\cdot\omega_g]_+. \tag{4.10}$$

Chronology has not been dropped in this formula.

Lemma 4.4 (Simultaneous simplex enlargement)**.** *If $S_a = \{s < t_1 < \cdots < t_a < s+\tau\}$ and $a \ge 1$, then*

$$|S_a - [0,\Theta]\mathbf{1}| = \frac{\tau^a}{a!} + \Theta\frac{\tau^{a-1}}{(a-1)!} = \frac{\tau^a}{a!}\left(1+\frac{a\Theta}{\tau}\right). \tag{4.11}$$

The empty simplex has zero-dimensional volume one.

Proof. Fix the ordered gaps $t_k - t_1$. The first-time length is $\tau - (t_a - t_1)$ in S_a and increases by Θ in the enlargement. Integrating over the gap simplex gives the original volume and Θ times its $(a-1)$-dimensional gap volume, proving the formula. For $a = 1$ there are no gap coordinates and the length is $\tau + \Theta$. □

Since $\mathcal{D}$ is a product of slab merger simplices,

$$|\mathcal{D}_\Theta| \le \prod_i \frac{\tau^{a_i}}{a_i!}(1+a_i) \le (1+n)^L\prod_i\frac{\tau^{a_i}}{a_i!}, \qquad \Theta = \tau. \tag{4.12}$$

This also covers no-merger graphs. A partial final slab can be enlarged to its full assigned slab before the estimate.

4.5 Prefix flux sum, factorials, and normalization

Keep (v, y, t, ω) fixed in (4.10) and sum over the discrete tree of successive merger-pair choices. At a node the preceding choices and all intervening redundant slots have already been processed. Its incoming velocities are common to every possible next pair; no future pair determines a past reflection. Prefix failure has weight zero. On the energy bin $E_0 \le P$, (4.4) gives for the slab-i lines

$$\sum_{p<q}[(v_p^{\mathrm{cur}} - v_q^{\mathrm{cur}})\cdot\omega]_+ \le \sum_{p<q}|v_p^{\mathrm{cur}} - v_q^{\mathrm{cur}}| \le (n_i - 1)\sum_p |v_p^{\mathrm{cur}}| \le n_i^{3/2}\sqrt{P}. \tag{4.13}$$

A child failing an existence or strict-order test contributes zero. Induction bounds the flux products, with $R_G^{\mathrm{ext}}Q_G$ retained, by $\prod_i (n_i^{3/2}\sqrt{P})^{a_i}$. Normal integration costs only a fixed constant per merger. Only after this prefix bound is it legitimate to discard $R_G^{\mathrm{ext}}Q_G \le 1$ for the positive time-volume bound (4.12). No nonchronological velocity history is used in the induction.

The coefficients (4.2) were fixed by the finite forest exploration before the physical change of variables. Neither the shear nor positive enlargement introduces fresh labels, another cluster partition, or a new order permutation. H06's fresh-label count bounds all coefficients apart from ordinary merger-pair choices by

$$C_K(1+n)^{\gamma_K}C^a\prod_i\frac{1}{n_i!}.$$

The $O_K(1)$ redundant labels, initial-link vertices, and crossing roots cost fixed polynomials. Marking an incident C-slot costs at most $n + O_K(1)$ additional choices.

Integrating the initial-link cylinders contributes

$$C_K \varepsilon^{2p} (1 + E_0)^{p/2},$$

because (4.9) includes both its $\varepsilon^2 \tau |u_l|$ and ε^3 terms, $p \le K$, and $|u_l| \le 2\sqrt{E_0}$. In the connected case

$$\mu^{m-h} \varepsilon^{2(m-1-p)} \varepsilon^{2p} = \mu^{1-h}. \tag{4.14}$$

The original extra ε per physical initial link is lost, but the connected root normalization is preserved. This also covers a connected graph with no dynamic merger, where $p = m - 1$.

4.6 The enlarged summation closes

Apply the fixed reserve (3.2)–(3.3). The velocity integral of $e^{-\beta_* E_0/4}$ is C^m and the energy sum is

$$\sum_{P=1,2,4,\dots} e^{-\beta_* P/4} P^{a/2} \le C^{a+1} (1 + a)^{a/2}. \tag{4.15}$$

This follows by comparison with a gamma integral and Stirling's inequality; the lowest bin uses $P^{a/2} = 1$. Its base depends on β_*, not on K, p, or the moment order. The numerical base need not equal the old unmodified one; its independence of these later parameters is what matters.

Up to fixed polynomials, the remaining size expression is H's

$$C^{n+a} \tau^a (1 + a)^{a/2} \prod_i \frac{n_i^{3a_i/2}}{a_i! n_i!}. \tag{4.16}$$

Put $\nu_i = n_i - a_i \ge 0$; (2.12) gives $\sum_i \nu_i = O_K(1)$. Factorials imply

$$\frac{n_i^{3a_i/2}}{a_i! n_i!} \le e^{2n_i} (1 + n_i)^{\nu_i} n_i^{-n_i/2}.$$

Indeed replace $1/a_i!$ by $(1 + n_i)^{\nu_i}/n_i!$ and use $n_i! \ge (n_i/e)^{n_i}$, dropping further nonpositive powers. Zero counts contribute one. Also

$$\tau^a \le \tau^n \max(1, \tau^{-1})^{O_K(1)}, \qquad (1 + a)^{a/2} \le C_K (1 + n)^{O_K(1)} n^{n/2},$$

and convexity of $u \log u$ gives $\prod_i n_i^{n_i} \ge (n/L)^n$. Combining (4.3)–(4.16) proves the bound

$$\frac{|J|}{\tau} C_K (1 + n)^{\gamma_K + L + 1} (C\tau\sqrt{L})^n \mu^{1-h}. \tag{4.17}$$

For fixed L, the number of size vectors with total n is polynomial in n. Section 3 chose $C\tau\sqrt{L} = CT/\sqrt{L} < 1$ independently of that polynomial degree. Summing (4.17) proves Lemma 4.1. The marked contact never acquired an independent time coordinate; its unit-speed motion came from one collective shear.

5 Moving pulses and increment cumulants

Fix real $\phi \in C_c^\infty$ and write $M_\phi = \|\phi\|_\infty$ and $D_\phi = \|v \cdot \nabla_x \phi\|_\infty$. These are finite. For $0 \le s < t \le T$, put $\delta = t - s$ and

$$S_{s,t} = \sum_i [\phi(\hat{z}_i(t)) - \phi(\hat{z}_i(s))].$$

This is H's scalar record with pulses $-\phi$ at s, $+\phi$ at t, always ending at the same full time T.

Proposition 5.1. *For every fixed integer $j \ge 1$ and every $q > 0$, uniformly in s, t,*

$$|\kappa_j(S_{s,t})| \le C_j \mu \delta + C_{j,q} \varepsilon^q. \tag{5.1}$$

Consequently, for every assigned $q > 0$,

$$\mathbb{E}|\widehat{X}_t(\phi) - \widehat{X}_s(\phi)|^4 \le C(\delta^2 + \delta/\mu) + C_q \varepsilon^q, \tag{5.2}$$

$$\mathbb{E}|S_{s,t}| \le C\mu\delta + C_q \varepsilon^q. \tag{5.3}$$

Proof. On each actual retained history a designated line with no displayed C-atom in $(s, t]$ flies freely there. Refining every autonomous C-cluster by its complete ordered accepted history, as in H04, makes this statement literal. O-atoms change no velocity. Its scalar mark therefore satisfies the positive majorant

$$|y_\ell| \le D_\phi \delta + 2M_\phi N_{(s,t]}(\ell). \tag{5.4}$$

Only powers of the single final scalar record are used to test the actual cumulant. The two endpoint states in this inequality are a pointwise history argument, not an illicit two-record test of an endpoint measure.

Apply (2.4). For $h = |\pi| \le j$, use the retained connected expansion, its high-complexity tail, and the initial-link expansion (2.11). On the physical-link part, all other record factors are bounded by fixed powers of $2M_\phi$; one factor is bounded by (5.4). The unmarked absolute mass is $C\mu^{1-h}$ by the source summation, and the window term is $C\mu^{1-h}\delta$ by Lemma 4.1. The factor μ^h in (2.4) gives $C\mu\delta$.

We justify a single accuracy choice uniform in both pulses. Fix j and the desired power q. Keep exactly these two pulses and at most j designated labels to the fixed endpoint T. Their update maps have $|y| \le 2M_\phi$ uniformly in s, t. Compatibility and acceptance do not use the mark. In absolute estimates, remove pulse-to-segment selection restrictions; the possible placements cost only a fixed polynomial in size, or a cutoff power in the coarse regime. The stopped kernels and crude TV operator depend only on the test supremum. Thus the positive kernel proofs in Section 2.3 provide one majorant for all $(s, t) \in [0, T]^2$, not just constants for each fixed observation list.

Choose a coarse accuracy $A > q + 2j + 2$, then one K making (2.10) at most ε^A for all $h \le j$. Choose a tail target $q_{\text{tail}} > q + 2j + 2$ and then $D > 2K + q_{\text{tail}}$. Multiplication by μ^h leaves $O(\varepsilon^{A-2j})$ for the coarse part and $O(\varepsilon^{q_{\text{tail}}-2})$ for an initial tail $O(\varepsilon^{q_{\text{tail}}}\mu^{1-h})$. The unweighted variation $e^{-c\Lambda_L}$ beats every fixed power. These estimates use only bounded powers of y, or $|y|$, with no Gaussian weight for a generic microscopic remainder. Higher assigned powers first absorb every finite cross product below. None of these choices changes L or the dynamics. This proves (5.1).

The centered fourth-moment identity gives

$$\mathbb{E}|\widehat{X}_t - \widehat{X}_s|^4 = \mu^{-2}(\kappa_4(S_{s,t}) + 3\kappa_2(S_{s,t})^2).$$

Insert the preceding estimates and relabel the arbitrary error power, using $\delta \le T$, to get (5.2). For (5.3), positivity of the actual one-particle measure $c_1 = F_1$ gives $\mathbb{E}|S_{s,t}| \le \mathbb{E}\sum_i |y_i| = \mu c_1(|y|)$. Its same connected decomposition and (5.4) prove the bound. If $\phi = 0$ all statements are immediate. □

The remainder is an arbitrary assigned power, not a qualitative cycle remainder. This distinction is needed when passing to microscopic scales.

6 A final-slab history flag

We now derive the additional finite-kernel operation needed for a microscopic modulus. Let $(i-1)\tau \le a < b \le i\tau$ and stop the hierarchy at b. For a designated particle p put

$$r_p = \mathbf{1}_{\{\text{its velocity jumps in } (a,b]\}}.$$

This is a bounded measurable function of its final-slab history; it is not asserted to be recoverable from its endpoint.

Lemma 6.1 (Decorated output)**.** *On the output space $(z_p(b), r_p)$, the factorial hierarchy of the same pasted dynamics has the retained connected-diagram expansion and the unweighted TV, coarse, and initial-tail decomposition of Section 2.3, uniformly in a, b and the partial endpoint. For each fixed h and arbitrary $q > 0$ its actual decorated cumulant satisfies*

$$\left|\mu^h c_{h,b}^{\varepsilon,\mathrm{dec}}(r_1 \cdots r_h)\right| \le C_h \mu(b-a) + C_{h,q}\varepsilon^q. \tag{6.1}$$

Proof. In (2.6) replace the final delta, for each fixed autonomous partition, by

$$\delta_{((z_p^{\Psi_{\mathcal{C}}}(b), r_p(\Psi_{\mathcal{C}}|_{(a,b]})))_{p\in S}}. \tag{6.2}$$

Use the identical replacement in the positive stopped kernel. The caps give finite accepted histories, so the flag is measurable on the source's common full-measure history space. Apply (2.5) to bounded measurable tests of this new output. Its finite Janossy/forest proof still cancels spectators pointwise before integration; (2.8) still leaves $\mu^j/j!$. Thus the one-slab identity holds as an identity of measures on the decorated space, not only after endpoint projection.

Earlier slabs carry no jump flag. Compose their ordinary unmarked kernels with this final decorated kernel. In the final slab keep all designated histories explicit. A final independent singleton may be denoted $g_{i,b}^{\mathrm{dec}}$; it is the decorated one-root output on independent ordinary lower inputs. It is not the ordinary unmarked g_i followed by an attempt to recover its flag. Lower one-root factors can retain their canonical ordinary resummations.

For coarse bookkeeping, the finite one-root definitions can be opened along the at most h designated physical lines down to zero, with a constant record if needed. This is the source's record-preserving finite regrouping and projection operation. It duplicates neither an input nor an interface birth. Any needed bottom continuation can be supplied with at most h source dummy straight-root links at fixed cost $\varepsilon^{-h/81}$, together with the allowed top-root deficit $\varepsilon^{-C_* h}$. Their fixed powers and complexity are absorbed by the subsequent K; they are removed before the low-complexity physical-link estimate. The flag remains a final-output function. No record or line is introduced per collision.

Decorated outputs factor over disjoint global components because each root flag depends only on its own component history. Expand lower independent sums finitely for this algebra. Assigning fresh labels to components yields the same multinomial identity as in H04, Lemma `lem:diagram-labels`. Consequently

$$\widehat{F}_{b,S}^{\mathrm{dec}} = \sum_{\pi\in\mathcal{P}(S)} \bigotimes_{V\in\pi} D_{b,V}^{\mathrm{dec}}. \tag{6.3}$$

Finite inversion keeps exactly the connected decorated diagrams. An independent decorated singleton contributes only to order one; it cannot create a connection at order two or higher. Its order-one history integrals are final-slab diagrams with Gaussian Bol-bounded ordinary births, to which the same window estimate applies.

The variation argument must take place before the new pushforward. At the level of each positive history kernel, (6.2) preserves its mass. We do not lift a TV estimate of an unmarked endpoint measure to its histories. Rather, for any bounded decorated test H_0,

$$|R_s^{\mathrm{dec}}(H_0)| \le \|H_0\|_\infty \, (\mathcal{E}_{i,s}F_{i-1})(1).$$

In the recursion $\Delta_i^{\mathrm{dec}} = R_i^{\mathrm{dec}} + \mathcal{L}_i^{\mathrm{dec}}\Delta_{i-1}$, the lower error is the ordinary unmarked one. The decorated kernel's crude TV operator bound is unchanged: its positive partition/forest majorant sees only the supremum of H_0. The source stopped-variation induction and its partial-final-slab bounds therefore apply. The fixed bookkeeping losses above have the allowed form already absorbed in that induction.

The actual normalized decorated factorial mass is $\mu^{-|S|}\mathbb{E}(N)_{|S|} \le 1$, and restricting by flags only lowers it. The retained decorated measure has variation at most $1 + e^{-c\Lambda_i}$ at fixed orders. In the difference of the actual and retained finite partition polynomials every term has a variation-error factor; the other factors have these bounded masses. Thus the cumulant variation error is still $C_h e^{-c\Lambda_i}$.

For the high-complexity tail, the positive coarse integrand depends on a, b only through a multiplier at most one and a shorter final time domain. There are only L final slab indices. The same fixed root losses and (2.10) give one K for every a, b at a fixed assigned accuracy. Choose the initial-tail sharpness afterwards, as in Section 5. All of these are unweighted TV statements.

Finally refine each autonomous C-cluster by its complete ordered accepted C history, as in H04's one-root bridge and stopped-variation proof. The root velocity changes only at displayed C-atoms, so on each physical retained history

$$r_p \le N_{(a,b]}(p). \tag{6.4}$$

Bound all other flags by one and apply Lemma 4.1, after removing artificial coarse-estimate links. The normalization μ^h gives $C\mu(b-a)$. Choose the unweighted error powers before that multiplication to obtain the actual remainder $C_{h,q}\varepsilon^q$ in (6.1). □

Restricting the decorated output to $r = 1$ commutes with tensor products in (6.3). It is thus the ordinary factorial hierarchy of the point process retaining just the jumping particles. The lemma is a separate consequence of the finite history kernel, not an application of the additive-pulse theorem to a nonadditive endpoint mark.

7 A microscopic modulus

For a real path Y write $w(Y,u) = \sup_{|t-s|\le u} |Y_t - Y_s|$. Put

$$\alpha = \mu^{-3/2}. \tag{7.1}$$

Proposition 7.1. *For every real compact smooth ϕ, $w(\widehat{X}^\varepsilon(\phi), 2\alpha) \to 0$ in probability.*

Proof. Let $I_k = (k\alpha, (k+3)\alpha] \cap (0,T]$, for $0 \le k \le \lceil T/\alpha \rceil$. Any pair $s < t$ with $t - s \le 2\alpha$ lies in the closure of I_k for $k = \lfloor s/\alpha \rfloor$. For small ε, $3\alpha < \tau$; split I_k at its possible one interface into at most two pieces I_k^ι inside individual slabs. Define

$$\mathrm{Count}(I) = \#\{p : \hat{v}_p \text{ jumps in } I\} = \sum_p r_p.$$

A particle with no jump in $(k\alpha, t]$ flies freely; its test change between s, t is at most $3\alpha D_\phi$. Otherwise use $2M_\phi$. Pathwise,

$$\sup_{s,t\in\overline{I_k}}\left|\sum_p[\phi(\hat{z}_p(t))-\phi(\hat{z}_p(s))]\right| \le 3\alpha D_\phi N + 2M_\phi \sum_\iota \mathrm{Count}(I_k^\iota). \tag{7.2}$$

For root cardinality $h = 1, 2, 3$, the factorial cumulants of this restricted point process are those in Lemma 6.1. For $|I| \le 3\alpha$ they obey the *same actual error convention*

$$|\mu^h c_h^{\varepsilon,\mathrm{dec}}(r^{\otimes h})| \le C_h\mu\alpha + C_{h,q}\varepsilon^q.$$

Here h denotes block cardinality, d remains the test-vector dimension, and b the slab endpoint; no μ^h is inserted again in this already-normalized remainder. The exact factorial moment identity and the diagonal identity are

$$\mathbb{E}(\mathrm{Count}(I))_j = \sum_{\pi\in\mathcal{P}([j])}\prod_{V\in\pi}\mu^{|V|}c_{|V|}^{\varepsilon,\mathrm{dec}}(r^{\otimes|V|}), \quad j = 1, 2, 3, \tag{7.3}$$

$$\mathrm{Count}(I)^3 = (\mathrm{Count}(I))_3 + 3(\mathrm{Count}(I))_2 + \mathrm{Count}(I). \tag{7.4}$$

Since $x = \mu\alpha = \mu^{-1/2} \le 1$ and $\varepsilon^q \le 1$, every finite product is bounded by $C_j(x + \varepsilon^q)$, increasing the assigned source accuracy first if necessary. Thus

$$\mathbb{E}\,\mathrm{Count}(I)^3 \le C(\mu\alpha + \varepsilon^q). \tag{7.5}$$

All falling-factorial diagonal terms have been kept.

Markov's inequality and a union over $O(T/\alpha)$ pieces, with threshold $\eta\sqrt{\mu}$, give

$$\begin{aligned}\mathbb{P}\Big(\max_{k,\iota}\mathrm{Count}(I_k^\iota) > \eta\sqrt{\mu}\Big) &\le \frac{C(T/\alpha)(\mu\alpha+\varepsilon^q)}{\eta^3\mu^{3/2}} \le C\eta^{-3}(\mu^{-1/2}+\varepsilon^q)\\ &\le C\eta^{-3}(\mu^{-1/2}+\mu^3\varepsilon^q) \longrightarrow 0, \qquad q > 6.\end{aligned} \tag{7.6}$$

The last, weaker form uses a common error choice convenient for chaining; the first form aligns directly with the actual error in (6.1). Since $\mu^3\varepsilon^q = \varepsilon^{q-6}$, $q > 6$ suffices. A second-moment union based on its leading term would give only $(T/\alpha)(\mu\alpha)/(\eta^2\mu) = T/\eta^2$, which does not vanish.

The free-flight term in (7.2), after division by $\sqrt{\mu}$, is $3D_\phi\mu^{-1}(N/\mu) \to 0$ in probability by (2.3). For exact centering, (5.3) gives uniformly for $|t - s| \le 2\alpha$

$$\mu^{-1/2}|\mathbb{E}S_{s,t}| \le C\mu^{1/2}\alpha + C_q\varepsilon^q = C\mu^{-1} + o(1).$$

All intervals are half-open $(a, b]$. The phase paths are right-continuous; a jump at the left endpoint is already reflected in its endpoint state, and a right-endpoint jump belongs to the count. Pasting creates no interface jump. Thus these conventions cover grid endpoints and interface splits. The three bounds prove the proposition. □

8 Chaining, continuity, and exact-centering transfer

Lemma 8.1 (Chaining to the microscopic scale)**.** *Let Y^ε be real cadlag processes on $[0, T]$, with Y_0^ε tight, and suppose*

$$\mathbb{E}|Y_t^\varepsilon - Y_s^\varepsilon|^4 \le C(|t-s|^2 + |t-s|/\mu) + R_\varepsilon, \qquad R_\varepsilon\alpha^{-2} \to 0.$$

If $w(Y^\varepsilon, 2\alpha) \to 0$ in probability for (7.1), then Y^ε is C-tight in the J_1 Skorokhod space.

Proof. Use the chaining exponent $\vartheta = 1/8$, distinct from the shift parameter θ. Let $t_{k,j} = jT2^{-k}$ and let k_ε be the largest k with $T2^{-k} \geq \alpha$. The finest spacing lies in $[\alpha, 2\alpha)$. For $Z_k = \max_j |Y_{t_{k,j+1}} - Y_{t_{k,j}}|$ the union bound gives

$$\mathbb{P}(Z_k > 2^{-k\vartheta}) \leq 2^k \{C(T^2 2^{-2k} + T2^{-k}/\mu) + R_\varepsilon\} 2^{4k\vartheta}.$$

Summing from k_0 to k_ε gives three terms bounded respectively by

$$C \sum_{k \geq k_0} 2^{-k/2}, \qquad C\mu^{-1}(T/\alpha)^{1/2} = C\mu^{-1/4}, \qquad CR_\varepsilon(T/\alpha)^{3/2} \leq CR_\varepsilon \alpha^{-2}. \tag{8.1}$$

On the complementary event, successive dyadic approximations give for nearby finest-grid points a bound $Z_{k_0} + 2\sum_{k>k_0} Z_k \leq C_\vartheta 2^{-k_0\vartheta}$. For arbitrary s, t with $|t - s| \leq T2^{-k_0-1}$ pass to adjacent finest-grid points; each endpoint error is bounded by $w(Y, 2\alpha)$. For sufficiently fine terminal grids the possible fixed-level edge adjustments are absorbed in C_ϑ. Hence

$$w(Y, T2^{-k_0-1}) \leq C_\vartheta 2^{-k_0\vartheta} + 2w(Y, 2\alpha)$$

outside the exceptional probabilities in (8.1). First let $\varepsilon \to 0$ and then $k_0 \to \infty$. Thus, for each $\eta > 0$,

$$\lim_{u \downarrow 0} \limsup_{\varepsilon \to 0} \mathbb{P}\big(w(Y^\varepsilon, u) > \eta\big) = 0.$$

To verify also the supremum-tightness hypothesis of [5, Proposition VI.3.26], partition $[0, t]$ into intervals of length at most a fixed $u > 0$. Deterministically,

$$\sup_{t \leq T} |Y_t^\varepsilon| \leq |Y_0^\varepsilon| + \lceil T/u \rceil \, w(Y^\varepsilon, u).$$

Given a probability tolerance, the double limit first supplies a fixed u for which $w(Y^\varepsilon, u) \leq 1$ outside a set of that tolerance in the $\limsup_{\varepsilon \to 0}$. Tight initial values then supply a finite bound for $|Y_0^\varepsilon|$ at the same tolerance. With this u fixed, the displayed inequality proves asymptotic tightness of the suprema. Together with the continuous-modulus limit, this gives C-tightness by the cited criterion. This is standard tightness machinery; small individual jumps alone would not prove it. □

For clarity we reproduce the Gaussian continuity and uniform-centering deductions already established in antecedent notes to this work. They are not new claims here; the accompanying provenance identifies the archived notes and their precise sections. H02 constructs the target on

$$E_a^0 = \{h \in C(\mathbb{R}^6) : e^{-a|v|^2} h \in C_0(\mathbb{R}^6)\}, \qquad \|h\|_a = \sup e^{-a|v|^2} |h|,$$

with bounded maps to $L^2(\Omega)$ for $a < \beta$, uniformly in time. For compact smooth ϕ, $r \mapsto A_r\phi$ is continuous and bounded in E_a^0 for $a > 0$. The source weak identity and field/source bounds give

$$\zeta_t(\phi) = \zeta_0(\phi) + \int_0^t \zeta_r(A_r\phi)\, dr + W_t(\phi), \qquad \mathbb{E} \int_0^T |\zeta_r(A_r\phi)|^2\, dr < \infty. \tag{8.2}$$

The integrand is strongly measurable in L^2 by the source propagator construction. Its indefinite integral has continuous paths by Cauchy–Schwarz. The Gaussian noise has $\mathbb{E}|W_t(\phi) - W_s(\phi)|^4 \leq C|t - s|^2$, so Kolmogorov yields a continuous modification. Equation (8.2) gives the continuous version of each $\zeta(\phi)$, with all joint covariances unchanged. A finite intersection of probability-one events handles a finite list.

Proof of Theorem 1.2. For each coordinate, (5.2) has $R_\varepsilon = C_q\varepsilon^q$ with arbitrary $q > 6$, so $R_\varepsilon\alpha^{-2} = C_q\varepsilon^{q-6} \to 0$. Proposition 7.1 and Lemma 8.1 give C-tightness of the pasted coordinate. Its initial value has the same initial ensemble as the true flow and is tight by H's initial finite-dimensional theorem. For a fixed vector, its continuous modulus is bounded by the sum of the coordinate moduli, and its initial vector is tight. It is therefore C-tight.

Let $S_t = \sum_i \phi(z_i(t))$ and $\widehat{S}_t = \sum_i \phi(\hat{z}_i(t))$. On the coupling (2.16), simultaneously in all times,

$$\sup_{t\le T} |S_t - \widehat{S}_t| \le 2M_\phi N \mathbf{1}_{E_\varepsilon}.$$

Hölder's inequality with the fixed 22nd moment of N/μ gives

$$\begin{aligned}\sup_{t\le T} \frac{|\mathbb{E}S_t - \mathbb{E}\widehat{S}_t|}{\sqrt{\mu}} &\le C\varepsilon^{-1}\mathbb{P}(E_\varepsilon)^{21/22} \\ &\le C\varepsilon^{-1+(21/20)(21/22)} = C\varepsilon^{1/440} \longrightarrow 0. \end{aligned} \tag{8.3}$$

The coupling exponent is $\eta_0 = 1/20$; it is distinct from the arbitrary probability threshold η in (7.6). Off E_ε the uncentered paths coincide at every time and $\mathbb{P}(E_\varepsilon) \to 0$. Thus

$$\left\| X^\varepsilon(\phi) - \widehat{X}^\varepsilon(\phi) \right\|_\infty \to 0 \quad \text{in probability.} \tag{8.4}$$

This works for a fixed finite vector. The identity time change bounds its J_1 distance by the supremum distance. The modulus inequality transfers C-tightness. No fourth-moment bound for the true flow restricted to the exceptional event is asserted or required.

For each fixed ε, particle number is finite almost surely and $\mathbb{E}N < \infty$. Bounded-test sums are cadlag; their expectations are cadlag by dominated convergence using $M_\phi N$. The centered vector is therefore a random element of the usual J_1 space, whose Borel sigma-field is generated by coordinate evaluations. Every subsequential limit of the C-tight family is supported on continuous paths. Evaluation at any time is continuous there, so H's joint finite-dimensional convergence identifies its finite-dimensional laws with those of (1.8). Finite-dimensional laws determine the law on $C([0,T],\mathbb{R}^d)$, giving a unique subsequential law and convergence of the whole sequence.

On a Skorokhod representation of a J_1-convergent subsequence with continuous limit, paths converge uniformly almost surely. This is a consequence on that representation, not weak convergence in the nonseparable uniform-topology Borel space on all cadlag paths. □

9 Scope, provenance, and supporting verification

The conclusion retains actual unsmoothed trajectories, exact microscopic centering, the whole-space ensemble including $N = 0$, and the entire supplied regular interval. It does not assert a distribution-valued path limit beyond every fixed finite test vector, regularity of a Boltzmann solution beyond the assumed interval, or a quantitative process-convergence rate. The auxiliary cutoff coupling is not independent resampling at slab interfaces.

The essential new passage is (4.1): a determined redundant contact gains a window factor through a collective direction, with strict prefix chronology, linked-coordinate cylinders, and the original factorial normalization. The decorated-output extension is proved from the finite kernel rather than assumed from an endpoint record theorem. The cumulant, third-moment, chaining, Gaussian continuity, and uniform exact-centering steps are credited machinery or deductions once those interfaces are paid. The antecedent notes established continuity and uniform-centering transfer and identified the missing window estimate. These inherited deductions are reproduced here and are not claimed as separate new results.

The accompanying source map distinguishes H's stipulated inputs, the inherited arguments, and the new lemmas, and identifies the antecedent notes. No worldwide-priority claim is made.

Supporting scripts check exact finite rational countercharts, chronological and initial-link shears, energy and scaling identities, the structural O-tie rejection, factorial diagonals, and exponent arithmetic. Such calculations can detect a sign or convention error, but do not prove the general history sum, kernel remainder bounds, or stochastic limit. Their bytes, provenance, and reproduced outputs are recorded alongside this paper.

No full Lean certificate exists for this theorem. A separately compiled companion contains nine supporting declarations: elastic-pair momentum and squared-norm energy in a real inner product space; four affine time-shift identities; finite all-line energy under a pair replacement and its concrete elastic update; and the rational exponent budget. Their exact types and compilation/axiom receipts are recorded in the accompanying provenance. They do not prove covariance of the entering-root construction, invariance of strict chronology, a Jacobian or measure-preservation theorem, existence of all-time histories, the window estimate, the decorated kernel bound, or the path limit. The full geometric and probabilistic argument remains prose mathematics; these finite identities are not additional research results. No unrelated or pending external formalization is used as a proof premise.

Computational assistance. OpenAI Codex and other AI research/review agents assisted with the written argument, independent checks, and typesetting. The package identifies the scope of those written reviews and finite computations. No additional human author, affiliation, external peer review, or formal proof certificate is implied.

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[4] Y. Deng, Z. Hani, and X. Ma, *Long time derivation of the Boltzmann equation from hard sphere dynamics*, arXiv:2408.07818v3, July 18, 2025. The regular-interval framework is imported through H; Section 1.4.3 is a comparison concerning fluctuations.

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